TITLE: Interfacing of Multiplex 7 Segment Display.

AIM:

This experiment includes the interfacing of LEDs and Seven segment display with 8051 Microcontroller.

After completion of this experiment, students will be able to

1) Interface an 8051 microcontroller with a display device and can perform the desired task.
2) Program a 8051 microcontroller using Embedded C Language

THEORY:

The 7-segment display consists of seven LEDs arranged in a rectangular fashion. Each of the seven LEDs is called a segment because when illuminated the segment forms part of a numerical digit (both Decimal and Hex) to be displayed. An additional 8th LED is sometimes used within the same package which is the indication of a decimal point (DP), when two or more 7-segment displays are connected together numbers greater than ten can be displayed.
So by forward biasing the appropriate pins of the LED segments in a particular order, some segments will be glowing and others will remain as it is, allowing the desired character pattern of the number to be generated on the display. This then allows us to display each of the ten decimal digits 0 to 9 on the same 7-segment display.

Now accordingly terminals are taken common, so there are two types of display:

1. Common Cathode display
2. Common Anode display

1. The Common Cathode (CC) –

In the common cathode display, all the cathode connections of the LED segments are joined together to logic “0” or ground. The individual segments are illuminated by application of a “HIGH”, or logic “1” signal via a current limiting resistor to forward bias the individual Anode terminals (a-g).

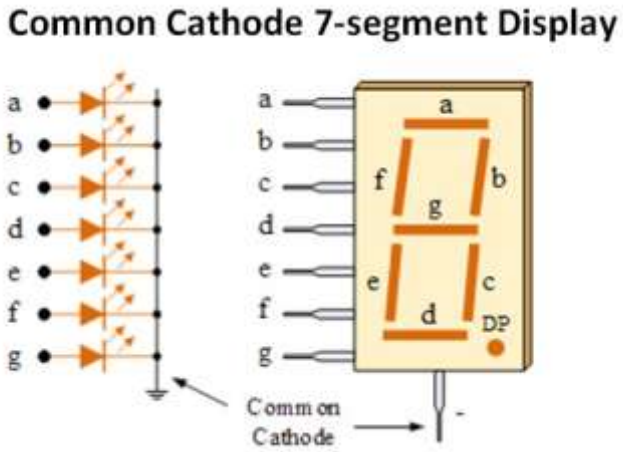


Fig.:- Common cathode Terminology

2. The Common Anode (CA) –

In the common anode display, all the anode connections of the LED segments are joined together to logic "1". The individual segments are illuminated by applying a ground, logic "0" or "LOW" signal via a current limiting resistor to the Cathode of the particular segment (a-g).

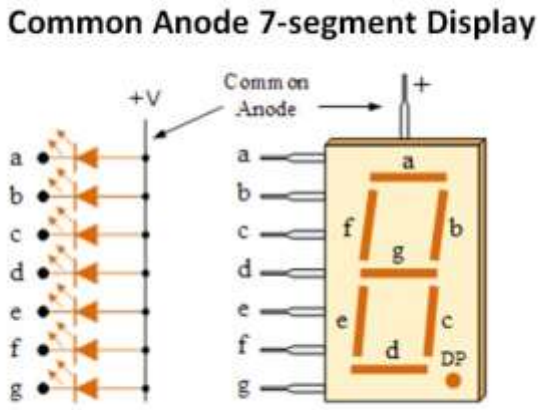
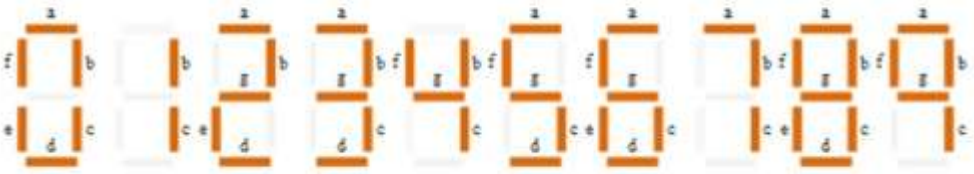


Fig.:- Common Anode Terminology

In general, common anode displays are more popular as many logic circuits can sink more current than they can source. Also note that a common cathode display is not a direct replacement in a circuit for a common anode display and vice versa, as it is the same as connecting the LEDs in reverse, and hence light emission will not take place. Depending upon the decimal digit to be displayed, the particular set of LEDs is forward biased. For instance, to display the numerical digit 0, we will need to light up six of the LED segments

corresponding to a, b, c, d, e and f. Then the various digits from 0 through 9 can be displayed using a 7-segment display as shown below.

7-Segment Display Segments for all Numbers.



Seven segment Display shows numbers

Refer the table given below to know the hex values for displaying a specific number on the 7-segment display during the simulation.

Decimal	ABCDEFG	A	B	C	D	E	F	G
0	0×7E	1	1	1	1	1	1	0
1	0x30	0	1	1	0	0	0	0
2	0x6D	1	1	0	1	1	0	1
3	0x79	1	1	1	1	0	0	1
4	0x33	0	1	1	0	0	1	1
5	0x5B	1	0	1	1	0	1	1
6	0x5F	1	0	1	1	1	1	1
7	0x70	1	1	1	0	0	0	0
8	0x7F	1	1	1	1	1	1	1
9	0x7B	1	1	1	1	0	1	1

Conclusions:

QUESTIONS:

1. What is the difference between two types of seven segment display?
2. Explain the Port and its pin with 8051 Microcontroller.
3. Write a C Code display 0 to 9 along with A to F on 7 segment display.